



**Department of Computer Science and Engineering,
PES University, Bangalore, India**

**Lecture Notes
Problem Solving With C
UE24CS151B**

***Lecture #3
Problem solving using File handling functions***

**By,
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Unit #: 3**Unit Name: File Handling and Portable Programming****Topic: Problem solving using File handling Functions**

Course objectives: The objective(s) of this course is to make students

- Acquire knowledge on how to solve relevant and logical problems using computing Machine.
- Map algorithmic solutions to relevant features of C programming language constructs.
- Gain knowledge about C constructs and its associated ecosystem.
- Appreciate and gain knowledge about the issues with C Standards and it's respective behaviours.

Course outcomes: At the end of the course, the student will be able to:

- Understand and Apply algorithmic solutions to counting problems using appropriate C Constructs.
- Understand, Analyze and Apply sorting and Searching techniques.
- Understand, Analyze and Apply text processing and string manipulation methods using Arrays, Pointers and functions.
- Understand user defined type creation and implement the same using C structures, unions and other ways by reading and storing the data in secondary systems which are portable.

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Solve the below programs using File handling functions.

Link for Solution:

https://drive.google.com/file/d/1zak9bA_bak_wuu40iDBN_a2c5TVfUL1g/view?usp=drive_link

Data files: <https://drive.google.com/drive/folders/1nBeirvXZcvWr7N7gk0-CPWq7LuQ5ffbE?usp=sharing>

Level-1: Banana

1. Aryan wants to leave a short message for his friend in a file, so that his friend can read it later. Write a C program that **asks the user to enter a message (one line) and saves this message to a file named "message.txt"**. Display the content of message.txt on the terminal.
2. Ria has a file filled with mixed content—**letters, symbols, spaces, and numbers**. She wants to access the file content and **display only the digits (0–9)** from the file, ignoring everything else. Help Ria by writing a program that reads the file and prints all the digits present in it.
3. Arjun wants to write a C program that **reads a text file line by line, removes the newline character, appends a colon(:) at the end of each line, and displays the updated lines on the screen**. Help him implement this using file handling.

Input file:

Arjun likes coding
He practices every day
C is his favorite language

Expected output:

Updated lines:
Arjun likes coding:He practices every day:C is his favorite language:

4. Naina is a librarian at a small town library. Every evening, she updates a file with the list of books issued that day. She writes a C program to do the following:
 - **Accept the book title and the borrower's name from the user.**
 - **Write this information to a file named "issued_books.txt" in the format: Book: <title>, Borrower: <name>**
 - **Repeat this until the user types "stop" as the book title.**
5. Anvi maintains a small grocery shop. Every day, she enters the **details of items sold into a text file named sales.txt**. Each line in the file contains **Item name (string), Quantity sold (integer) and Price per item (float)**. Calculate the total sales amount for the day using a C language.

Level-2: Orange

6. Priya likes writing her thoughts in a digital diary every evening. At the end of each month, she wants to know how much she has written — **the number of characters, words, and lines** — **so she can track her writing habits.**

You are a computer science student helping Priya to read her diary file named diary.txt and print the total number of characters, total number of words and total number of lines.

7. Rahul is a huge fan of the Indian Premier League (IPL) and wants to analyze match statistics from past seasons. He has a **CSV file named matches.csv that contains data about all IPL matches.** Each row in the file contains details like **match ID, season, city and many more details.** Rahul is particularly interested in the inaugural season (2008). Help him write a C program to read the file and **count how many matches were played in the year 2008.**

You can assume that the file matches.csv is properly formatted. The second column in each row contains the match date in the format YYYY.

8. Anjali is a volunteer at her school library. She wants to build a small C program that **helps to record and save book information into a file. Each book has the following details - Title (string), Author (string) and Year of Publication (integer).** Write a C program that:

- Defines a structure Book with the above fields.
- Allows the user to enter the details of **N books.**
- Writes all the book details into a file named books.txt.
- Then, read the file and display the book details on the screen.

Level-3: Jackfruit

9. In the heart of TechSpark Robotics Lab, Thakshini, a brilliant computer science student was always finding clever ways to solve real-world problems. Her **project involved training humanoid robots to handle college administrative tasks like managing entry at tech fests and workshops.** These robots were deployed temporarily, but every evening, the lab manager, Mr. Rajan, found himself stuck - “I just can’t keep track of which robot worked today,” he sighed, flipping through a dog-eared notebook filled with scratched-out names and IDs. He asked “Thakshini, can your smart bots log themselves in next time?” That joke sparked Thakshini’s next mini project. She decided to make a text-based terminal program in C that:

- **Took the robot’s ID and name from the user.**
- **Stored them in a plain text file called employee.txt.**
- **Allowed Mr. Rajan to view all logged robots with just a few clicks.**

She wants to create a menu-based system:

1. Add Robot Staff
2. View Robot Staff
3. Exit

Sample execution:

===== ROBOT STAFF CONTROL PANEL =====

1. Add Robot Staff
2. View Robot Staff
3. Exit

Enter your choice: 1

Enter the number of robot staff to log: 2

Enter Robot ID: 1001

Enter Robot Name: Xena

Enter Robot ID: 1002

Enter Robot Name: Bolt

Robot records added successfully!

===== ROBOT STAFF CONTROL PANEL =====

1. Add Robot Staff
2. View Robot Staff
3. Exit

Enter your choice: 2

--- LOGGED ROBOT STAFF DETAILS ---

Robot ID: 1001

Name: Xena

Robot ID: 1002

Name: Bolt

10. Meena, a lab assistant in the computer science department, is developing a simple login system for the lab's admin panel. **Only authorized users are allowed to change system configurations or access student records.** To begin with, she hard-codes the registered username as "student". Whenever a user tries to log in, the system should verify if the typed username matches. **To enhance the system further, Meena decides to** store all login attempts (both successful and failed) **in a file named login_log.txt**. Every time a user attempts to log in, the program should:

- Prompt the user to enter a username.
- Compare the entered username with the valid one ("student").
- Display a message: "Login successful" or "Access denied".
- **Append the attempt** and its result (Success or Failure) to login_log.txt along with the entered username.

Help Meena write a C program that performs this task using **file handling**.

A test of brilliance for exceptional students

ABC is a growing retail store that wants to digitize its product inventory and customer billing using simple C programs that work with .csv files. The system must track products, generate bills, maintain stock levels, and create sales reports.

Product data is stored in a file named products.csv:

Format -> ProductID (unique), ProductName, Category, Price, Stock

Sale data is stored in a file named sales.csv:

Format -> BillNo (unique to each purchase), ProductID (same as in products.csv), QuantitySold, TotalAmount

Banana:

1. Take product details (name, category, price, stock) from the user and add them to products.csv. Assume that the ProductID of a new product will be one more than the previous ProductID (100, 101, 102...).
2. Read all entries from products.csv and display them in a well-formatted table.
3. Take a product name as input and search for that product in products.csv. Display its details if found, otherwise display an appropriate message.
4. Use sales.csv to find the total amount made in sales so far, as well as the average bill value for each sale.
5. List every product category and the number of products in each.

Orange:

1. Record a purchase by taking product IDs and quantities for each product bought as input, and update the sales.csv and products.csv files accordingly. Include edge cases such as when the quantity trying to be bought exceeds the current stock of the product or when the stock of a product reaches zero after a purchase. Note: Multiple products being bought at once will all have the same bill number.
2. Sort all products by price and write the sorted list to a new file called sorted_products.csv.
3. Delete a product from products.csv using its product ID.
4. Add a stock alert feature that prints an alert message directly to the console when the stock of any given product goes below 10 units.

Jackfruit:

1. Identify and print the top 3 products that bring in the most revenue for company ABC, along with how much revenue they generated.
2. Implement a rewind function that takes n (number of sales) as a parameter, and undoes the effects of the last n transactions on both products.csv and sales.csv.

Data lives on, beyond your code — with files !!